The trick for getting reliable structured output from an LLM

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The trick for getting reliable structured output from an LLM

SWIPE ->

Getting an LLM to return valid structured data is harder than it looks.

Asking nicely for JSON works about 90% of the time. In production, the other 10% is what pages you at 2am.

Here's the approach I used in Tank:

Instead of parsing free text, I gave the model a hidden tool. Think of it as a "final answer" tool whose input schema is the output format I want.

The model doesn't "write JSON." It calls a tool.

Then Pydantic validates the arguments. If they're wrong, the validation error goes back to the model as a tool result, and it corrects itself on the next turn.

Why this works:

- Models are heavily trained to call tools correctly
- Validation is code, not hope
- Failures become a retry loop instead of a crash

It's not magic. Add a retry cap so a stubborn model can't loop forever.

Have you found a more reliable approach for structured output? I'd like to compare notes.

#LLM #AI #Python #Pydantic

Thanks for reading.

Follow **Yasir Usman** for more on AI, ML and computer vision.

Code: github.com/yasirusman85/Tank